Cosmic Queries – Planet 9

Podcast: Startalk Podcast

AI Executive Summary

Core Thesis This episode of *StarTalk Cosmic Queries* serves as a wide-ranging exploration of fundamental astronomical and physical principles—ranging from planetary classification and galactic evolution to the nature of dark matter and space technology—aiming to clarify scientific mysteries through the lens of observational evidence, modern physics, and the mechanics of the universe.

Key Arguments & Data * **The Search for Planet Nine:** Scientific suspicion regarding a large object (Planet X) in the outer solar system is driven by gravitational anomalies observed in Kuiper Belt objects; their movements cannot be explained solely by the Sun's gravity, suggesting a massive, distant source of gravity. * **Criteria for Planetary Status:** Pluto was demoted to "dwarf planet" because it fails two of the three International Astronomical Union (IAU) criteria: it has not "cleared its neighborhood" of other objects and its orbit crosses the path of Neptune. * **Galaxy Morphology & Evolution:** The shape of a galaxy is determined by the rotation speed of its progenitor gas cloud. * **Elliptical Galaxies:** Formed from low-rotation collapses; they are efficient star producers that have largely depleted their gas. * **Spiral Galaxies:** Formed from higher-rotation collapses; they remain gas-rich and serve as "stellar nurseries." * **Irregular/Disturbed Galaxies:** Often the result of galactic collisions, where gravity pulls stars into "train wreck" patterns without direct physical stellar collisions. * **Quantum Physics & Computing:** The modern IT revolution and the development of semiconductors are entirely dependent on quantum physics; understanding the behavior of electrons within atoms and molecules is essential to controlling electricity in integrated circuits. * **Dark Matter & Black Holes:** Because dark matter interacts via gravity, black holes will inevitably absorb it via their event horizons. Once inside, the dark matter cannot escape, regardless of its lack of electromagnetic coupling. * **ISS De-orbiting Projections:** The International Space Station is projected to de-orbit around 2030–2031; due to the vastness of the Pacific Ocean, it remains the primary target for uncontrolled re-entry debris. * **Atmospheric Protection:** Earth's atmosphere serves as a critical shield

against small meteorites, burning up most debris before it reaches the surface.

Actionable Takeaways * For Self-Directed Learners: To build a structured foundation in astrophysics, begin with "Cliff Notes" style summaries (e.g., *Astrophysics for People in a Hurry*) and progress to comprehensive textbooks such as *Welcome to the Universe*. * **For Scientific Literacy:** Recognize that scientific classifications (like the status of Pluto) are not arbitrary but are responses to new observational data provided by advancing technology (e.g., the Keck Telescope). * **For Policy Awareness:** Monitor NASA funding and international space competition (specifically regarding China's rising space capabilities), as these political drivers directly impact the lifespan and modernization of orbital infrastructure like the ISS.

Full Transcript

Chuck, another query is grab bag. Yes. My favorite one is the person wants to know what the chance of getting hit by a meteor. Yes, and you were actually taken out by a meteor. Almost. Oh, OK. Dillman 11 on Star Talk. Welcome to Star Talk, your place in the universe where science and pop culture collide. Star Talk begins right now. This is StarTalk Cosmic Queries. And as always, it's a grab bag edition. And I got Chuck nice with me. How are you doing, Chuck? Hey, Neil. Doing well, man. Thanks for asking. There's no shortage of people's question. Yes. People really inquiring minds want to know. So bring it on. Let's jump right into it. Andy L says, hello, stellar intellects and also cohost of the week.

My name, he has assumed that the co-host is clearly not of a stellar intellect. So this is. But he's allowing it to have been other co-hosts because we have other co-hosts to come in. Yes, yes, yes. That's why I'm not taking it personally. He says, Hey, my name is pronounced Andy. See, now that lets me know that he thinks it's me. See, now you're full of it. And D. And I know what the D stands for too. Anyway. By the way, that's another name. It's Nick name for Richard. Thank you. He says it's also a nickname for private detective. That's correct. Private Dick. That's right. Yes. And that's why in the opening song to the movie Shaft, Isaac Hayes sings, who's the black private dick that's a sex machine to all the chicks?

Yeah, he was talking about invest investigations That was that was a line I had to like listen to a few times and then I have to look up the definition and one of the definitions of dick was private detective So that was a weird right with a scripting there. Okay, go ahead way to go eyes the case He said I heard there may be a large planet or dwarf planet orbiting beyond Pluto Planet 9. The so-called Planet X. What exactly are the scientific observations causing this suspicion? Could the new Rubin telescope find such an object? Thank you, Andy, from Thousand Oaks, California. I do not know the targeted zones of the Rubin telescope. But generally, if you're going to find a planet and you don't know that it's a planet, you need several observations because otherwise the stars, though they move to orbit the galaxy, are so far away from night to night, they basically keep the same

position. But something orbiting the sun would not, from night to night, week to week, month to month. So I think, in principle, the Rubin telescope would be able to detect such a thing. But it would have to be directed to a region of space where we think it is and then hang out there just for that purpose. But that's not what the telescope was designed for. It was designed as a survey telescope of the entire sky looking for whatever moves in the survey zone. So that's my first point. Second, the way we have colleagues who think about this and do these experiments, the Pluto we discovered in the last 20 years, 25 years, is just the most prominent member of a new swath of real estate discovered in our solar system called the Kuiper Belt, named after a Dutch, I think Dutch American astronomer, George Kuiper by mid-century. He hypothesized the existence of a repository of comets that didn't participate in the formation of the rest of the solar system. In fact, he made the inverse prediction. He would say, He said there's a region out there where there would be comets if they didn't participate. Okay. It was one of those kind of backhanded predictions. Turns out there are thousands of dirty ice balls orbiting well beyond Neptune, including Pluto. Okay. So, and the Kuiper Belt is huge. So, observations of Kuiper Belt objects tell us that maybe they are responding to another source of gravity in the outer solar system and not just in their orbit around the Sun. There's movements that can't otherwise be explained if they're simply orbiting the Sun. So that allows you to say maybe there's another source of gravity out there that's mysteriously tugging on these objects we do see. Now, if it's a planet, It's going to be much bigger than just a Kuiper belt object, but if you can't see it, it must be much farther away. So they're placing it at a distance below their detection limits because they're saying otherwise we would have seen it. And when you do this, this thing is really far away. So there's still a lot of speculation. There's been some skepticisms of the observations and the interpretation, but that's what it is. And that's how we infer that there might be a planet nine.

or a new Planet X, because we all know Pluto got demoted. Great, yes. So there you have it. And Andy crawled through a mile of it anyway. Okay. This is Eric Leroy who says, greetings, Neil and Chuck Eric Leroy here, exercise physiologist and space flight physiology researcher from NYC. And my question is, what are the requirements established by the IAU that led to the demotion of Pluto as one of the major planets of our solar system?

In my opinion, they are not specific enough and should include granular measurements in addition to the phrase clearing its neighborhood. I believe if Pluto was given its title as the ninth planet back, there would be a surge in scientific and astronomical interests around the world of all ages. Thank you for all that you do in the name of science. So he's saying, let's create a controversy that will spur interest in and astrology once again, astronomy once again, I said astrology just, oh Jesus, astronomy once again by giving Pluto planetary status back. The rules weren't diabolically conceived just to get rid of Pluto. It's not how that worked. Okay. So here's what happened. Let me remind people, as our telescopes got better, As the Keck telescope started looking in the outer solar system, in the region of space where Pluto orbits, we started finding other objects. And there was an object whose size rivaled that of Pluto. And no one was prepared to call that a planet. So that forced a whole fresh look at what Pluto is, where it is, how it's orbiting, and why. Undoing so, we found, not only is Pluto more than half ice by volume, like comets are, Pluto is orbiting a region of the solar system that we would discover has countless thousands of other icy bodies in the outer solar system.

These, we would more commonly call comets. Pluto was the first discovered because it was the biggest and the brightest. You would expect the biggest and the brightest to be discovered first. Then we have better telescopes and now we're popping up, we've lost count of the number of Kuiper Belt objects that are out there. So this is not different from the year 1801. where a new planet was discovered orbiting between Mars and Jupiter. That was a big gap anyway between Mars and Jupiter on a planetary scale. We said, something has to be there. Let's keep looking. We found a planet. We called it Ceres, named in the tradition of a Roman god, as were all the other planets. Then we found another one, and then another, and then another. Ceres, palace, P-A-L-L-A-S, Vesta, and Juno. Wow. And he says, wow, the planet count just. Look at all these planets. A big leap. And then we said, well, wait a minute, how come when I look at them through a telescope, you don't see a disk the way you see a disk like you see Jupiter has you see a weather bands on it, Saturn, you see rings. You know, that's Pluto calling you right now. When you look at the Other planets, you see the physical, spherical object that is the planet. When you look at these planets, they're just dots of light. They're really tiny. And we know they're

orbiting the sun and they're reflecting sunlight because we can follow the orbit. But when you look at stars, they're just dots of light. But we know these are not stars. They're just star-like. Wait a minute.

What's a word for star like? I'm going to say asteroid. Asteroid. Once we started discovering dozens of these things, we said, we can't call these things planets. What instead has happened is that we have discovered a brand new swath of real estate in the solar system that we didn't previously know was there. Another belt. A belt. And generally, it's a belt if it's flat but goes around the circle, like it rims the sun, the asteroid belt. So overnight, those four new planets were, quote, demoted to asteroid status. And now there's countless hundreds of thousands, likely millions of asteroids discovered in catalog orbiting in that zone. Okay. Having discovered Pluto first and thinking it's a planet, and then discovering another object and another. And we kind of been down that road before. We have countless Kuiper Belt objects all in an orbit that is spread around the sun, enclosing the trajectory of Pluto. That's the first strike against it. It's not isolated in its orbital zone. It is one among many whose size rivals it. Wow. Okay. Two.

Yeah, it's small but that alone doesn't indict you. It's big enough to be a sphere. If you're small, it's got enough mass to be a sphere. If you have much less mass than that, then the gravity doesn't overcome the structural integrity of the rocks and the rocks went out. So most asteroids look like Idaho potatoes. Okay, there's whatever the rock is, that's its shape. But when you get bigger, gravity pulls the high places into the low fills them in and you become a sphere, eventually. Interesting fact about gravitational physics. All right, so Pluto did not clear its orbit. Pluto, because it's in a swath of real estate completely filled with other objects that rival it in size. What else do you have to be to be a planet? You have to be big enough to be round. Pluto is big enough to be round.

It is. But it needs more than one criterion to be satisfied for planet status. Having failed one of those two, it got, quote, demoted to dwarf planet status. However, you know, Ceres, the first asteroid discovered? That's the only asteroid that's round. Of course, it was big enough for gravity to make it round. And of course, that's going to be the first asteroid you discover. Ceres has more mass than all the other asteroids combined. So Ceres got a

promotion. At the same time Pluto was demoted to dwarf planet, asteroid Ceres was elevated to dwarf planet. And so Pluto has company. There are many dwarf planets in the solar system to join it. And like I said, It's half ice by volume. If you brought it to where Earth is right now, heat from the sun would evaporate the ice and it would grow a tail. And the third criterion I left out, sorry, you can't cross the orbit of other planets. Oh, that's amazing. Who doesn't spend 20 years out of the 48 year life closer to the sun than planet Neptune?

It crosses Neptune. It goes to the outside, to the inside. To the inside. And by the way, that's what's going to do that routinely. They cross all the planet orbits as they plunge in towards the sun. Right. So out of three criteria, Pluto failed two of them. It's a fraud! Pluto! You're a fraud! However, while it had planetary status, we discovered a new element It basically invented a new element in the laboratory that we named after Pluto. And so Pluto now has an element on the periodic table from false pretenses. Just thought I put it out there. And in sequence, there's element 92, which is uranium, named after which planet? Take a guess. I'm going to say Mars. No, of course, Uranus. Uranus. And then there's Neptunium.

This element number 93, named after Neptune and element 94, plutonium, named after Pluto. Pluto was discovered in 1930. Plutonium was isolated in 1940. It was weaponized, turned into a bomb and used in warfare five years later. Damn. Damn. That's right. Wow. Oh, look at that. Pluto. the most destructive non-planet effort. Wow. That was the first time I've seen you give that exhaustive list of Pluto's, Pluto's failure to be a planet. Um, I, I, well, listen, Eric, I hope that did it for you, man. Cause. Get over it. That's terrible. But really get over it. All right. Hello, I'm Thank You Broke Allen and I support Star Talk on Patreon. This is Star Talk with Nailed Grass Tyson. Let's go to Nirav Shah who says hello Dr. Tyson, Lord Nice. I am from India now living in Arizona.

I have listened to you, Dr. Tyson, for years and appreciate your storytelling such that it helps me recall things and I share it with my fellows. Could you please share the connection between theory of relativity, quantum physics and semiconductor physics, how the former eventually allowed the latter to develop. Thank you for everything that you guys at StarTalk do. There's no

direct connection that I know of between relativity and what happens in a circuit board. But relativity is what we call modern physics, which was discovered in the 20th century just as quantum physics was. So often, if you take a college class in modern physics, you learn both relativity and quantum physics together in a first introduction to, in your first step beyond classical physics. All right, so quantum physics, quantum, people say you took a quantum leap.

A quantum is a very small unit discovered originally as a small unit of energy that cannot be continuously varied, a quanta of energy. You can't have one and a half quanta. You can have one, you can have two, you can have three, you can't have fractional quanta. That's the whole construct of that understanding. So in circuits, you're moving electrons around. To understand electrons and what they do and who they behave and why you need quantum physics You need quantum physics to understand anything that's going on inside of molecules and atoms and nuclei So you don't have to single out the semiconductor and its importance to civilization being fed by innovations in quantum physics because it's all quantum physics So, I don't know how to reply other than to say just a couple of things about semiconductors. A semiconductor typically uses a wafer of silicon which is an insulator mixed with some other material that's a conductor of electricity. Insulator does not transmit it, conductor does. Semiconductor is configured so that you can control the passage of electricity.

at will and on command depending on the needs of the circuit. And so semiconductors became highly useful, especially in computing. And the race was how to make the tiniest semiconductor that had the most number of circuits in it. So you can, when I grew up, computers were the size of whole rooms. That's how old I am, right? Those were four function calculators. walk into the room and it was hot because it had vacuum tubes that radiated heat. Quantum physics enables the IT revolution entirely. And so I don't know what more to say about that other than that we're in the centennial decade of quantum physics. And let's celebrate that fact because civilization as we know it would not exist without it.

Well, thank you quantum Yeah, there you go. All right, let's move on to Derek Evans Hi, dr. Tyson Lorde nice Derek here from South Windsor,

Connecticut My question is about the shape of galaxies as we know they take many forms outside the stereotypical spiral shape But what ultimately dictates a galaxy's shape? Do they all rotate and if so does shape indicate? age or developmental stage? That sounds very psychotherapy. What is your developmental stage? Are you lagging behind? I'm a little stunted, you know what I mean? All the other galaxies seem to have it together, man. You know, I'm just really confused all the time. And I'm so angry all the time too. I don't know if it's because I'm a spiral galaxy or a sombrero galaxy or I honestly don't know what my identity is. It's a galaxy named the sombrero galaxy. It's not a kind of galaxy. It's the name of the galaxy. Oh, is that the name? Yeah, it's like a sombrero. And it look okay. There you go. Okay. So if you start out with a gas cloud. Right. Huge gas cloud.

Depending on how fast it's rotating, if it's hardly rotating at all and it collapses, it doesn't flatten. It stays kind of three-dimensionally globular. We call them elliptical galaxies. There's spherical elliptical shapes. Okay. And they rotate very slowly. Okay. All right. If you have more rotation than that, you will flatten. as you collapse. And you become very flat, flatter than a flapjack. You become as flat as a crepe, almost as flat as a tortilla. And so our galaxy, the Milky Way, when seen edge on would almost disappear from your sight line for that reason. And what happens is you, a galaxy gets stars from its gas clouds. And in this elliptical galaxies, For reasons not fully understood or were not fully understood, the last I looked at this problem, that's the more the safe way to say that. Elliptical guarantees were very efficient making stars. So they made all, they used up all their gas and made all the stars and there's no gas left. So when you don't have gas, you're not doing what? I'm going to say ruining the dinner because you're rude at the table. These are different gases Chuck.

Not hydrogen sulfide. So, yeah, so elliptical galaxies are mostly gas free. They made all their stars early on, and the stars are just aging right now. Whereas spiral galaxies, they retained 10%, 20%, sometimes 30% of their mass in gas. Wow. So there's still stars being formed. So when you see those beautiful photos, from JWST and Hubble, these gas clouds and stuff. We probe those gas clouds. It's where stars are born. Stellar nurseries. Yes, stellar nurseries. So these are the differences among the galaxies. Spiral galaxies are inefficient converting their gas into stars. Elliptical galaxies are

very efficient. And so there you have it. Now there are galaxies that are very disturbed looking. Those are the galaxies that are late. You come near me, I'll kill you. You think you're gonna collide with me? I will kill you. Try to collide with me and see what happens. Those are the belligerent category of galaxies.

Because it reminds me of just bullies. You don't even know the bully. You step on their toe. You do that again. I'll kill you. Exactly. It's like, bro, get in therapy. What is your problem? Galactic therapy. So in my lifetime, there was a catalog of galaxies that was published by a Caltech astrophysicist named Halton Arp, nicknamed Chip. Chip Arp. and it was called the Art Atlas of Peculiar Galaxies. He went around all the sky, the night sky, he catalogued galaxies that were just weird looking, disturbed looking, distorted looking. And we didn't have the answer to the question, did galaxies become this or were they born this way? The way, what's your name was, Lady Gaga. Mr. Gaga Galaxy. Were they born this way or did something happen to them? It was not until we were able to bring computing power to this problem in the late 60s, but mostly in the 1970s where we can say what happens to two grand design spiral galaxies if they collide? Oh my gosh, it's a train wreck.

Stars don't actually collide with each other, but they feel each other's gravity. And the gravity casts stars hither and yon. And it looks like a train wreck. So if you stop action, the footage of galaxies colliding, here they are coming near each other. Here they are partly overlapping. Here they are deeply overlapping. And you do this. Then you have a template handed to you by your computer. of colliding galaxies in all stages that they could collide. Now you hold that to Chip Arp's catalog, and you can explain 90% of those galaxies. I had a mentor once who said a crashed Lexus is not a different kind of Lexus. It's still a Lexus. Why he chose Lexus? I don't know, but that was his car brand that he used.

And there are also galaxies that are just irregular in shape for no good reason. You know what we call those? I'm going to go with irregular galaxies. Chuck, you're such a fast study here. So we have spiral galaxies, elliptical galaxies, irregular galaxies. They look kind of like puppy vomit on the ground. They tend to be very gaseous. which means they can make

stars. And then there are galaxies that are just generally small, a catch basin of galaxies. We call them dwarf galaxies. Oh, of course. Irregular galaxies are dwarf galaxies. You can also have small elliptical galaxies. We call those dwarf galaxies as well. You can see them going off to work singing hi-ho, hi-ho. I wrote an essay, one of my early writing forays, where I was paid.

I think it got paid 300 for this essay for a newsletter, and it's called The Galaxy in the Seven Dwarfs. That's what it was called. Because we have, our galaxy has at the time, seven known dwarf galaxies that orbit us. Typically, a dwarf galaxy is a member of some other association of galaxies. So there you have it. That's more galaxy evolution than maybe you bargained for, but that's how it goes. And Chuck, those are the basic categories of galaxies, but there are also galaxies that just do weird things that might be hybrids of these others that I just listed for you. There are quasars and blazars and lenticular galaxies and radio galaxies that give mostly radio waves. So there are variants on this. But what I described to you are 95 percent of all the galaxies in the universe.

Okay. Well, very cool. All right. Well, thanks a lot, Narav. Let's move on to Melinda Washington. She says, hello from North Carolina. As a new autodidact, I'm excited to explore astrophysics in a focus and structured way instead of jumping between topics. How would you recommend breaking the subject into manageable pieces? What materials would you suggest and in what order should I approach them to build a strong foundation. Thank you both for always sparking my curiosity and inspiring me to keep learning, always looking up Melinda Washington. Oh, love it. Love it. So at the risk of sounding self-serving, I co-authored a textbook with two colleagues at Princeton University when we co-taught a class. One of us taught Stars and planets, the others taught galaxies and the third taught cosmology. The class started out with 30 people and over the years it had hundreds and hundreds. This was at Princeton University. We said, why don't we turn this into a textbook? We did. The textbook is highly readable. It's not like, learn this for the problem set and the syllabus.

And so the book is called Welcome to the Universe. And if you get it through the museum's website, the museum shop, I think it's shop.amnh.org.

I think they sell it there. If you get it, it's pre-signed by me. I pre-signed the books that they sell with my fancy overpriced purple ink bottle filled fountain pen. If you don't want to jump into a textbook, There's a tiny version of it that is the, the Cliff Notes version of it is called a brief welcome to the universe. So what you might do is start with that. That'll wet your appetite and then you graduate to the full up textbook. And it's, you know, it's 400 pages. It's a full up textbook, but it's very readable. You can sit on the couch and, and, uh, and get lost. It's a page turner. Yeah. I have one in my bathroom, Melinda.

That's what I recommend. And it's very sequenced and very pedagogical, I should say. Very nice. All right. Well, that's very, that's, that's excellent. And don't read my bestselling book, Astrophysics for People in a Hurry. Astrophysics for People in a Hurry? Because clearly you're not in a hurry. A, B, what's in that book is me going through an entire curriculum and cherry picking the coolest, most mind-blowing things. and putting it into one volume. So I skip over stuff that you don't even know would have been there, but in a full-up curriculum in a college course, it belongs there. But if I just want to wet people's appetite, that's what I did. Cherry picking the coolest things in the universe for that book. Very nice. Yeah, I was talking to, I was on a show yesterday and doing an interview and the host read, astrophysics for people in a hurry. And they said, I don't care how, how much I'm not in a hurry. I still didn't get what he was talking about. So by the way, astrophysics is not astrophysics for nummies. Yeah. It's real astrophysics. It's real astrophysics. For its, its interest level.

So, yeah, don't think it's a lesser... Yeah, don't think you're going to get in there. And it's not going to be like, you know, this is a galaxy. See, galaxy spin. Spin, galaxy spin. It ain't doing that. Okay. So... All right. This is Christie Jiangu, or Jiangu. She says, Hi, Dr. Tyson and Lord Nice. This is Christie from Romania, writing to you from Italy. Where do we get all these international people? Wow. I love it. Man. She says, it is possible that dark matter is actually normal matter hidden in the guts of the space time fabric. I think she meant, is it possible? Because it's actually says it is, but is it possible? I'll read it that way.

Is it possible that dark matter is actually normal matter hidden in the guts of the space-time fabric? Imagine that on a space-time fabric there are vast regions so deformed like crumpled paper ball that it is impossible to see directly the matter inside those regions. Will that explain why we can't see or feel it but we can feel the gravitational effect? It's possible to verify this measure the time that light takes to travel through those regions of crumpled deformed fabric, since it has no cover or larger distance than what we actually see with our telescopes. Thank you. Keep looking up. Wow. I don't know that space time can hide things. That'd be kind of diabolical to science.

space time is hiding crucial information from the court, you know. So, I've imagined multiple times that maybe dark matter, which expresses gravity in our own space time, maybe dark matter is ordinary matter in a parallel universe that's similar in an idea to, you're tucking it away within folds of the space-time continuum. If it's from another universe, then it's ordinary gravity leaking into ours, forcing us to say, oh, what is this mysterious substance? We can't see it, taste it, but it's real in another dimension. And I look forward to knowing whether or not that's true, because it would be kind of cool if it was. I would love it. Because to me, that theory, not theory, but that postulate says that now we got to go ahead and start, you know, investigating other other universes. Yeah, except I don't want to be the first to visit them. The laws of physics are slightly different. You could just collapse into a pile of goo on the sidewalk. So yeah, be careful. Well, we'll send we'll send an advanced team.

Send a journal through. Yeah, exactly, you know. All right, this is Stephen Schultz who says, hello, StarTalk family. Stephen Schultz from Dayton, Ohio. Thank you all for taking the time to read my question. I've recently read that the ISS is scheduled to make its fall back to Earth in 2030 or 31. Will this be something you guys plan on watching in person? Will it be possible to see it from an island in the Pacific, assuming that's where it will land? Thank you guys for all of the amazing content. So is it going to de-orbit in 2030? Yes. Well, I think we'll probably keep it up there. You know, when you threat it to... to defund something, people come out of the woodwork and say, no, don't do that. Keep it going. And by the way, what a great way to get funding. Well, you can cut off this money, but when this

thing fall out the sky, you're on your own. Yeah. So the Pacific Ocean is the great space toilet bowl on Earth because the Pacific Ocean spans a third of all of Earth's longitude. It's huge.

so that you can drop something out of the sky without much precision, and you'll still hit the Pacific Ocean. Look at that. Can you watch it? Yeah, go take a rowboat out to the middle of Pacific Ocean when this happens, and there's no guarantee it will fall just over your head compared with other places. Your horizon in a rowboat might be 15 miles away, but re-entering spacecraft That horizon would be small compared to all the places it could land above your head. So, yeah, in principle, you could see it. Maybe we'd have drones out there filming it, but most of the space station will burn up in the atmosphere. Most of it. We expect that. And by the way, I'm not trying to go look at that because with my luck, it will fall on me. So I ain't trying. I like the idea of a drone filming it, though.

Keep in mind that the space station was built beginning in the early 1990s. So how many years ago? That's 30 years. If you visited someone's home and they had a television from the early 1990s, they would be the object of mockery to you. They'd also not be watching TV. You can't get anything on a TV from the early 90s. Technology has advanced since then. And if you sent up a space station today, it would probably look different or be much more modern than whatever we have got going up there now. So China plans to also have like a competing space station to our own. And that's why people are a little spooked in Congress that they would have the new high ground relative to what we have enjoyed over the decades. So the real issue here, Stephen, is to get more funding for NASA and space research because let them know that we're going to lose to China. And quite frankly, they will have the upper hand militarily. Yeah. That's all it takes. That's all it takes. And then everybody will just run with it. Yeah. There you go. This is Steven Skelton. And Steven Skelton says, Do we know the actual odds for someone, anyone to be hit by a small meteorite excluding large extinction level events? I find it amazing that only one person and Hodges has been recorded being struck by and no one else in at least 1,000 years of history has been recorded as being killed. With 8 billion people on earth, would you need to see odds on the magnitude of something like 1 trillion two, one, to help explain the lack of recorded strikes. Okay, so if we didn't have an

atmosphere, you'd be dead. I would need time to do a proper calculation on this. It's a population of the Earth, how much space we take up, what the surface area is, the fact that people aren't spread evenly around the world, that not all the meteors that strike Earth make it to Earth's surface, that they burn up.

I'm going to do a proper calculation and report back. But here's something I can tell you. I was almost hit by a meteor. Did I tell you about this? You of all people, if you got hit by a meteor, everybody would say, see, there is a God. The universe killed the universe guy. Well, what guy are you talking about? Right? Right. So, I think I've told this story before. Back in 1999, I was on the Brooklyn Bridge in the middle of the night, maybe one in the morning, in November for the Leonid Meteor Shower, which was on its 33-year cycle. where you expect not a meteor shower but a meteor storm. Comets leave debris that make meteor showers. And that debris orbits the Earth in front of and behind the comet. And Earth plows through that at the same time each year. That's why meteor showers repeat annually. All right.

This meteor shower is associated with a comet as a 33-year orbital period around the Sun. And if you're near where the comet had just passed the Sun or just come in front of the Sun, you get extra meteorite flux coming in. Because all the crap that falls off a comet is closer to where the actual comet is than scattered completely in its orbit. And all this debris continues to orbit the Sun. All right, I'm on the Brooklyn Bridge with my three year old daughter, three and a half year old daughter. I woke her up for this. Okay. We're there and I'm watching. We're getting a meteor every 90 seconds. I'm not quite storm level, but definitely satisfying. Okay. Then I'm looking up and I see a star in the sky just appear that it got brighter and brighter. I said, that's odd. I wonder what that is.

And then it disappeared in a puff of smoke. I said, oh my god! This was a meteor headed straight for me. But in the instant I saw it, I'm thinking, gee, I wonder what this is. I'm just being curious scientists until I put two and two together. And then I realized the day you die from a meteor strike is the day you are staring up at a light that gets brighter and brighter. And that'll be the last thing you saw in your life. So I almost hit, I think it would have hit me because I saw no streak left, right, or sideways. And then it just

simply disappeared. I was protected by Earth's atmosphere. Thank you, pollution. What? It's not pollution. No, the atmosphere protects you without the pollution. No. Oh, OK. That's true. We're good. We're good.

I checked. We only have time for one more. One more. What you got? All right. Here's the last question. It's Vail Stein. Dear Dr. Tyson and Lord Knice, my name is Vail and I'm from, oh wait. My name is Vail and I'm a seven year old math and science kid from Philly. Seven year old math and science kid. Okay. Wow. He says, do black holes absorb dark matter through the event horizon? And if so, with the black holes charge, mass and spin change? Or can dark matter flow in and out of black holes as it is not electromagnetic and only has some of the properties that matter does? All right. First of all, let me just say this. If you are from Philly, you damn sure ain't going to Philadelphia public schools.

There's a seven year old asking that question is not happening in a Philly public school. Damn, Chuck. No, I can't. That's my home tonight because you're from Philly. So you have, you have. I can make that joke. I can make that joke. All right. Anyway. All right. So it turns out the only property that dark matter has that we can interact with is gravity. It doesn't have electromagnetic coupling with any of our particles. That would mean it doesn't have electromagnetic forces, which are required to make molecules and hold atoms together. So it has gravity. And if it has gravity, we are left with no... skepticism over whether a black hole would just as happily absorb dark matter as it does regular matter. And the matter would not be able to come out any more than regular matter could because it's responding to gravitational forces. So you can bet that a black hole sucked in all the dark matter that was just in its area.

But if you're at a safe distance from a black hole, it's not a giant sucking machine all of a sudden. It'll pull in anything that was close. But if the sun became a black hole, for example, we would still orbit it. It would get very cold very quickly, but we would still orbit. We're not so close that we would get sucked in. And so there's an old saying, there's no such thing as gravity. Earth just sucks. Oh, wow. So, yeah, it's an excellent question. Wondering whether Dark Matter would have different properties in the vicinity of a black hole. And everything we know and understand about it tells us that it

wouldn't. There it is. Now, go back to your basement and be working on a super weapon to destroy the superhero. Exactly. Way to go, Vail. Nice, nice, nice work, nice work.

But that's all the time we have for it. So always good to have you there for these grab bags, Jack. Always a pleasure. Always a pleasure. All right. This has been Start Talk Cosmic Queries and another grab bag edition, which are always crowd pleaters. I am Neil deGrasse Tyson, your personal astrophysicist, as always, bidding you to keep looking up.